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Semester: 8th-B

Assignment : #02

Course: ML

Apply KNN Algorithm on Dataset.

Height (cm)	Weight (kg)	T- shirt-size
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Dataset Description:

158	58	M
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158	59	M
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158	63	M
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160	59	M
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160	60	M
-----	----	---

163	60	M
-----	----	---

163	61	M
-----	----	---

160	64	L
-----	----	---

163	64	L
-----	----	---

165	61	L
-----	----	---

165	62	L
-----	----	---

165	65	L
-----	----	---

168	62	L
-----	----	---

168	63	L
-----	----	---

168	66	L
-----	----	---

170	63	L
-----	----	---

170	64	L
-----	----	---

170	68	L
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The dataset contains three attributes.

1) Height (cm)

2) Weight (kg)

3) T-shirt size

→ In this dataset height and weight are the input features.

→ T-shirt size is the target class.

Query point:

A new person's T-shirt size needs to be predicted.

Height = 161cm

Weight = 61kg

Query point = (161, 61)

Distance formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

→ where x represents height.

→ y represents weight.

Distance calculations:

Distance with (158, 58)

$$d = \sqrt{(161 - 158)^2 + (61 - 58)^2}$$

$$d = \sqrt{3^2 + 3^2}$$

$$d = \sqrt{18}$$

$$d = 4.2$$

Distance with (158, 59)

$$d = \sqrt{(161 - 158)^2 + (61 - 59)^2}$$

$$d = \sqrt{3^2 + 2^2}$$

$$d = \sqrt{13} = 3.6$$

Distance with (158, 63)

$$d = \sqrt{(161 - 158)^2 + (61 - 63)^2}$$

$$d = \sqrt{3^2 + 2^2}$$

$$d = \sqrt{13} = 3.6$$

Distance with (160, 59)

$$d = \sqrt{3^2 + (-2)^2}$$

$$d = \sqrt{5}$$

$$d = 2.2$$

Distance with (160, 60)

$$d = \sqrt{1^2 + 1^2}$$

$$d = \sqrt{2} = 1.4$$

Distance with (163, 60)

$$d = \sqrt{(-2)^2 + 1^2}$$

$$d = \sqrt{5}$$

$$d = 2.2$$

Distance with (163, 61)

$$d = \sqrt{2^2 + (0)^2}$$

$$d = 2.0$$

distance with (160, 64)

$$d = \sqrt{1^2 + (-3)^2}$$

$$d = \sqrt{10} = 3.2$$

Sorted Distances:

Height	weight	size	Distance.
160	60	M	1.4
163	61	M	2.0
160	59	M	2.2
163	60	M	2.2
160	64	L	3.2
158	59	M	3.6
158	63	M	3.6
158	58	M	4.2

Choosing the value of k :

The value of $k = 3$ is selected.

The three nearest neighbours are:

- 1) (160, 60) $\rightarrow$ M
- 2) (163, 61) $\rightarrow$ M
- 3) (160, 59) $\rightarrow$ M

Since the majority class among the nearest neighbours is M, the predicted class for the query point (164, 61) is:

predicted T-shirt size = Medium (M).